# Rupali Bhatnagar

Senior Software Engineer

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## Work Experience

| ServiceNow | Sr. Software Engineer (Security Operations) | Mar'22 – May'25 |
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Individual contributor on Security Operations and Vulnerability Response, building on the ServiceNow© platform. Underlined features link to ServiceNow's public documentation.

- **Exception Management — Approval Rule Engine** for vulnerability exceptions — designed a multi-level, dynamic approval rule engine on ServiceNow Flow Designer supporting deferment and exception requests across differing enterprise policies; Flow Designer approvals shipped in **v15.0.2**. **Reduced approval time by 40%.**
- **Penetration Testing Request Catalog & Workflow Automation** — built the request catalog and automated intake, triage and routing for penetration-test requests across **50+ subsystems**, **cutting manual coordination by more than 30%** — the testing itself stays expert and manual by design.
- **Background Job Processing Framework** — architected the multithreaded framework for long-running tasks that previously strained system resources, partitioning work into parallel chunks, with migration via a configuration interface. Rollup jobs were converted to multithreaded background jobs in **v18.2.5**:
  - Multithreaded execution — **60% efficacy improvement** in some use cases, **1M+ tasks/month**.
  - Job status tracking and cancellation — **100% visibility** on executing and errored tasks.
  - Configurable **retry logic and dead-letter queuing** for failed jobs.
- **Real-Time Data Aggregation Engine** — high-performance aggregation framework for near real-time multi-dimensional analytics, powering key internal dashboards over **100K+ records in an 8-hour window**.

| ServiceNow | Software Engineer (Security Operations) | Apr'19 – Mar'22 |
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- **Assignment Rule Engine for Vulnerability Remediation** — cache-optimised engine prioritising infrastructure vulnerabilities and policy violations and routing **vulnerable items (VITs)** to the correct remediation groups via the Glide platform. Used daily by ITIL users; handles **1M+ vulnerable items daily**.
- **Assignment Recommendations on Predictive Intelligence** — classification models on the **Predictive Intelligence** framework with feature engineering and a reviewer-facing recommendation interface across **vulnerable items and vulnerability groups**; shipped in **v12.1.4**. **20% increase in task allocation accuracy.**
- **CISO & Vulnerability Dashboards** — led design and development of the CISO dashboard and supporting vulnerability dashboards, defining the key metrics and structuring the data pipeline. Used on a regular basis by **50+ organisations**.
- **Vulnerability Source Integration** — engineered integrations with **Veracode, Qualys, Tenable and Microsoft TVM**, building the **ETL pipelines** that standardise **500K+ vulnerability records/day** into a common model, with error handling and retry for reliable ingestion into downstream remediation workflows.
- **POA&M Generator** — auto-generated **Plans of Action & Milestones**, the remediation-plan artifact required under **NIST SP 800-53** and **FedRAMP**, directly from vulnerability findings rather than from control failures. Built at an internal hackathon and **adopted by the Now on Now support team**, who ran it against ServiceNow's own instance.
- **Hackathon Projects**:
  - **Diagnostic Framework** — internal support tooling that streamlined issue tracking and reproduction across customer instances, **reducing debugging time by 40%**.
  - **ML over Vulnerability Response data** — Predictive Intelligence **clustering** to group vulnerable items beyond the product's filter-based group rules, and **regression** to predict time-to-remediate, where shipped remediation targets are deterministic date arithmetic.
  - **High-Volume Data Generator** — scalable generation of **100M+ records across 20–30 tables** for load testing; used by the performance team.

| Darwinbox Digital Solutions | Software Developer | Aug'18 – Mar'19 |
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- Designed and implemented a microservice architecture capturing an **audit trail** of client data modifications, written in **PHP with the Yii framework**, listening to **MongoDB oplogs** and publishing to an **AWS SQS** queue.
- Built an **Oplog Processor in Node.js** diffing strings, arrays and embedded documents into human-readable audit entries, with an internal dashboard for adding trackable fields.

| NowFloats | Software Developer | Jun'17 – Jul'18 |
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- Primary work was the **Boost merchant portal's customer-relationship surface** — customer enquiries, call logs and call tracking — built in **C#/ASP.NET, Bootstrap, JavaScript, jQuery and Knockout.js** and serving **50,000+ merchant websites**.
- Integrated the **Facebook Graph API** so merchants could publish products and updates to their Facebook profile, page and shop (Ads extension).
- Built the **SPIDER** module using **machine learning** to predict sales revenue per representative from historical performance, with up to **84% accuracy**.
- Built an **offers and discounted-purchase flow** on **Kitsune**, NowFloats' serverless application framework, wiring in **k-Pay** and **Assured Purchase** — the escrow-backed payment-protection system that held buyer funds until an order was confirmed fulfilled — and surfaced it to merchants through the **Boost Keyboard**; also built the **Order Management** dashboard (**AngularJS, C#**) covering purchase, payment and delivery status and grievance redressal.
- Worked on the **appointment-scheduling system** across the Kitsune-built merchant websites — per-staff availability and slot modelling, booking and management — in **Node.js and C#**, generic across the clinic, salon, restaurant and hotel verticals, where it ships today as an appointment scheduler for practices and a room-reservation engine for hotels.

| **Finomena** (wound down 2017) | **Software Developer** | **Feb'17 – May'17** |
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- Built the internal dashboards for **Loan Management**, mandate (auto-debit) management, collections and transaction tracking on the small-ticket lending platform, in **PHP (Laravel)**, scaling to **8,000+ loan applications/month** with Eloquent ORM over MySQL.
- Built the **message categorisation service** feeding Finomena's alternative underwriting — the app read the borrower's SMS inbox and the service parsed and classified message content into structured credit signal, for a lender whose model scored borrowers with **no credit history**.
- Built the internal dashboard for the city **runner** fleet — the field agents handling collections and repayment follow-up.
- Used **Artisan** for task scheduling, cron handling and testing; Laravel modules for authorisation, validation, event handling and mailing.

| **Babajob.com** (acquired by Quikr, Jun'17) | **Software Engineer** | **May'16 – Jan'17** |
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- Built **TelephonySystem**, the **Golang** service behind the missed-call job channel and consumed by Babajob's C# platform — inbound and outbound call handlers against **Exotel** numbers, call-rotation strategy, incoming-call handling and the logging subsystem. **Stateful by design**: a first call created the jobseeker's profile, and later calls were recognised by source number and enriched it — so a feature-phone user built a profile by ringing a number. **10,000+ inbound calls and 20,000+ outbound SMS per day** on **RabbitMQ**, **Redis** and **MongoDB**; one of two developers on a five-person telephony team.
- Wired that profile into an **automated matching and application pipeline**: each completed call published the jobseeker profile to the outbound messaging service, which matched it against open roles through **Elasticsearch**, sent the shortlist by SMS, then walked the seeker through a sequential application that connected them to the job creator.
- Contributed to the **search subsystem in Java over Elasticsearch**, with dynamic query generation via Mustache templates.

## Selected Projects — 2025 to present

Independent and consulting engineering through **Tathya Dev** (tathya.dev) — **10+ production web products across 43 repositories and 4,238 commits in 12 months**, enabled by AI-augmented engineering workflows.

| **AI & agent tooling** | |
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| **propagate**<br>Claude Code plugin · Node, ESM | • Watches **~920 declared file couplings across 14 workspaces**, deriving drift from file *content* on demand rather than storing state — github.com/Rupali59/propagate. Hooks run in every agent session on the machine, delivering the matching known hazard before a risky command executes.<br>• Never edits a downstream file — it finds and explains, a person approves, the decision lands in an append-only ledger. Every detector ships a `selftest` asserting it can fire; auditing its own monitor to that standard surfaced **49 crashes in a stderr log nothing read**. The same rule runs through the state files it watches — each count names the command that derives it rather than the number, adopted after a header reading `8 edges · 8 verified` stood for eight days against a real 29, in the file warning about exactly that. **142 tests.** |
| **LLM Classification Pipeline**<br>Demand-intelligence ingestion · Gemini · schema-constrained | • Built a production LLM classification pipeline: Gemini under enforced response schemas at `temperature: 0`, batched, determinism from cache-on-first-write rather than the model, and a keyword-classifier fallback so a provider outage degrades quality instead of breaking ingestion.<br>• Model-proposed sources are **existence-checked against the live API before acceptance**; an inconclusive check flags rather than asserts. |

| **Products & platforms** | |
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| **Multi-tenant CRM & Workflow Platform**<br>Service-business CRM · Express · Next.js 14 · BullMQ | • Built an Express API and two Next.js 14 applications with a **database-driven RBAC/ACL engine**: runtime-editable roles, generic field-level access control over CRUD, per-actor tier projection, fail-closed reconciliation and a capability-matrix UI that shows the consequences of a change before an atomic save.<br>• WebAuthn + OTP, BullMQ/Redis queue, workflow engine, web-push. Six-phase engine cutover that deleted two legacy sweep passes only after verifying their outcomes equal to the new engine's and freezing them as a golden test — the cutover surfaced **five defects that existed only because the compiled graph had never executed**.<br>• Appointment-lifecycle scheduling in flight. Suite **104s → 28s**. **228 test files.** |
| **Content & SEO Platform**<br>Practitioner marketing site · Next.js 16 · TypeScript | • Shipped a production Next.js site on the computation service below.<br>• Nine templated landing pages plus a hub on a shared shell, `FAQPage` structured data with selective `noindex`, a server-side analytics pipeline, signed-cookie location handling with a TTL-indexed response cache, and public pages migrated to static generation. |
| **Practitioner Admin Console**<br>Scheduling, CRM and CMS · Next.js 16 · MongoDB | • Built the practitioner-facing console behind the public site — **~30 routes** over **26 Mongoose models**: schedule in day, week, month, list and waitlist views, a clients CRM with per-client kundli and birth-detail revert, tasks, notes, an articles CMS and fourteen settings and site-content editors.<br>• **WebAuthn passkeys** with Twilio OTP fallback and JWT sessions; practitioner notes **encrypted at rest with AES-256-GCM** under a documented dual-read key-rotation procedure. |

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| **Demand-Intelligence Platform**<br>Multi-source ingestion & scoring · Next.js · Mongoose · Gemini | • Built multi-source ingestion and opportunity scoring over Reddit, Google Trends, YouTube, Search Console and GA4, fed by the Gemini pipeline above, with an entity ontology mapping demand against supply.<br>• Search Console authenticated via **Workload Identity Federation** to remove service-account key fragility. Deployed; scheduled ingestion not yet cut over from the local agent. |
| **Ephemeris Computation Service**<br>Swiss Ephemeris over FastAPI · Python 3.12 · MongoDB | • Engineered a FastAPI service over Swiss Ephemeris, a native library with process-global mutable state that is **not thread-safe** — every call serialised through a single-worker executor that releases the event loop during computation.<br>• Astronomical-timeline subsystem: boundary and instant finders, lunar-month tables, per-year caching with TTL, and typed per-calculation endpoints replacing one overloaded route. **1,078 tests.** |
| **Motherboard**<br>API gateway & plugin platform · Go 1.25 · Next.js · Docker | • Architected a Go multi-module monorepo implementing an API gateway and multi-tenant plugin platform — **25 `go.work` modules, 652 tracked Go files**.<br>• Plugin services authenticate, publish routes to a client manager, and the gateway proxies behind a **fail-closed entitlement gate** and a plugin-key claim assertion. |
| **E-commerce Storefront & Embedded Admin**<br>Shopify · Liquid · Remix · Prisma | • Developed a Shopify Liquid theme and an embedded Remix admin: metafield/metaobject schema installer, headless catalogue seeders, and certificate verification implemented natively in Liquid rather than through an app proxy.<br>• Accessibility pass to a **44×44** touch-target floor. |
| **Production Portfolio Sites**<br>Filmmaker, designer, publisher · Next.js · React 19 · Playwright | • Shipped four production sites. A filmmaker portfolio with a Google Docs→works sync pipeline and multi-codec delivery: a hero codec ladder took a **9.9 MB** H.264 file to **2.3–2.9 MB** across AV1, HEVC and H.264 variants, one downloaded, suppressed entirely for non-watchers, with a **Level 4.0** re-encode because iOS silently refuses to autoplay above 4.2.<br>• A dual-track design portfolio gated by Playwright specs that fail on any hydration warning, plus axe WCAG 2.2 AA and text-spacing checks. A bilingual publishing platform with a separate Express admin carrying a Unicode-script language classifier that replaced an AI one on review. rtistic.vercel.app — Next.js 14, MongoDB, jest and CI, public source. |
| **Algorithmic Trading Platform**<br>Research and dashboard · Python · NATS · Next.js | • Built the research layer — edge estimators, a hypothesis framework and the data access beneath them, **1,000+ lines sole-authored** — and the **179-file** operator dashboard over it: strategy, system-health, market, trades, futures and options.<br>• Wrote a **design-drift ratchet**: each rule carries a **canary** and a selftest asserting it trips on a known-positive *and* stays silent on correct code — a lint rule that cannot fire is indistinguishable from a clean codebase. Adoption debt ratcheted to zero. |
| **Git Forensics & Recovery**<br>Incident work · git | • Recovered **154 commits** left unreachable in a live repository after a worktree removal combined with dropped stashes — **125 loose commits and 29 stashes** located, tagged under `refs/tags/rescue/*`, verified against `git fsck` and bundled. **8 of 29 stashes** held content present in no branch; one slice was reintegrated behind its tests and the rest deliberately left unrestored, each with a recorded reason.<br>• Separately diagnosed a repository whose `origin` pointed at an **unrelated project with no common ancestor** — 48 commits a single push would have sent to the wrong remote. |
| **Text Digitisation & Translation Pipeline**<br>Classical-text corpus · Sanskrit-texts · Python | • Built a **~4,800-line ingestion pipeline** for classical texts — a TEI/SARIT parser detecting division types by containment rather than frequency, a scraper splitting on the source's own headings, five verse-marker grammars detected per source, and per-work structural validators. IAST→Devanagari round-trip verified on **17,086 of 17,090** lines.<br>• One structural fix took depth errors in a single work from **3,522 to 6**. Publishes to a corpus of **92,166 verses across 56 texts**. |

## Dissertation & Coursework

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| **Dissertation** | Human Activity Recognition using an **HMM-based Intermediate Matching Kernel**, representing videos as sequences of sets of feature vectors. Re-implemented in 2026 as a containerised Python pipeline with a Flask front end — github.com/Rupali59/himk-implementation. |
| **Methodology** | Classifies a video into a predefined activity class. Feature extraction and classification in Python with scikit-learn, hmmlearn and OpenCV; originally prototyped in MATLAB with Piotr's CV Toolbox, PNetCDF and OpenMPI. |
| **ML Techniques** | Bayesian classifiers, Gaussian mixture models, perceptrons, artificial neural networks and support vector machines on linearly separable, non-linearly separable and overlapping data. |
| **Presentations** | Presented the components of an information retrieval system and Google's PageRank model at the departmental seminar on Information Retrieval. |
| **Other Projects** | Management Information System (MIS), Cybernetics Protector System, Product Management System. |

## Technical Skills

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| **Languages** | Go, TypeScript, JavaScript, Python, PHP, Java, C#, C++, GlidePlatform |
| **Web & Frameworks** | Next.js, React, Node.js, Express, Remix, Astro, Laravel, .NET, FastAPI, Tailwind, Bootstrap |

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| **AI & Agent Tooling** | Claude Code plugins, hooks and skills; structured LLM output and schema-constrained generation; eval harnesses and fixture-based regression suites; hallucination containment and graceful degradation; human-in-the-loop design; MCP |
| **Databases** | MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch |
| **Queues & Messaging** | RabbitMQ, BullMQ, AWS SQS, Redis Queue |
| **Cloud & DevOps** | AWS, Vercel, Fly.io, Docker, Shopify, Jenkins, Git, Linux |
| **ML** | scikit-learn, hmmlearn, OpenCV, MATLAB/Octave, pattern recognition and classification |

## Academic Profile

| Year | Degree | Discipline | Institution | Score |
|---|---|---|---|---|
| 2016 | **M.Tech** | Computer Science & Engineering<br>Pattern Recognition & Machine Learning | National Institute of Technology, Goa | 8.42 CGPA |
| 2014 | **B.E.** | Computer Science & Engineering<br>Graduated with Honours | LNCT, Bhopal (RGPV) | 8.58 CGPA |
| 2010 | XII (CBSE) | Science | St. Joseph's Convent School, Bhopal | 83.0% |
| 2008 | X (CBSE) | — | St. Joseph's Convent School, Bhopal | 88.0% |

## Certifications

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| **AI — 2026** | **Anthropic:** Claude Code in Action; Introduction to Claude Cowork; Claude Platform 101; Claude Code 101; Claude 101. **Google:** Gemini Enterprise Agent Ready; Create Your First Gemini Enterprise Application; Google Cloud Innovator; Developer Program Premium Tier; AI Fundamentals (Google Career Certificates). **Verify:** credly.com/users/rupali-bhatnagar-59 · g.dev/rupali59. |
| **Earlier** | WhatsApp API Expert, Wati (2024) · Ultimate JavaScript: Fundamentals, Code With Mosh (2025) · Neural Networks & Deep Learning, Coursera (2020) · JavaScript Algorithms & Data Structures, freeCodeCamp (2020) · Git Essential Training, LinkedIn Learning (2020) |

## Achievements & Activities

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| **Professional** | • Winner, NowFloats Hackathon for product innovation in sales — built the ML-powered SPIDER revenue-prediction module. 2018<br>• **Geektrust GOLD** — score **95.8** on the Metro Card backend challenge, with **Readability** and **Maintainability** assessed at **ADVANCED** tier by an independent reviewer. |
| **Academic** | • **Chancellor's Scholarship (Topper)**, Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal — awarded to the top 10 students statewide by theory marks. 2012, 2013<br>• **Silver Medal**, Persistent Systems Limited Programming Contest, inter-college. 2013 |
| **Other** | • Organised the intra-college programming competition, LNCT Bhopal. 2012<br>• Conducted cultural events, NIT Goa. 2014–2016 |

## Positions of Responsibility

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| **NIT Goa** | Teaching Assistant, Computer Science & Engineering Department | 2014–2016 |
| **NIT Goa** | Vice President, Computer Science Department | 2014–2016 |
| **LNCT, Bhopal** | Student Coordinator, Training & Placement Department | 2013–2014 |